

Question				Marking details	Marks available													
					AO1	AO2	AO3	Total	Maths	Prac								
3	(a)	(i)		- Can be used in {all organisms/cells/species}/ Can be used in both prokaryotes and eukaryotes (1) - For {most/ all} {reactions/ biological processes/ metabolic processes} (1)	2			2										
		(ii)		Glucose and water / accept correct chemical formulae (1) ADP and {P _i / phosphate} (1) <u>Contracted/ contracting</u> muscle (1)		3		3										
	(b)	(i)		A-chloroplast, B-mitochondria (1) Both correct for 1 mark		1		1										
		(ii)		<table><tr><th>Organelle A</th><th>Organelle B</th></tr><tr><td>thylakoid</td><td>Inner membrane (1)</td></tr><tr><td>Thylakoid space/lumen</td><td>Matrix (1)</td></tr><tr><td>stroma</td><td>Intermembrane space (1)</td></tr></table> 1 mark for each correct row Ecf from (i)	Organelle A	Organelle B	thylakoid	Inner membrane (1)	Thylakoid space/lumen	Matrix (1)	stroma	Intermembrane space (1)		3		3		
Organelle A	Organelle B																	
thylakoid	Inner membrane (1)																	
Thylakoid space/lumen	Matrix (1)																	
stroma	Intermembrane space (1)																	
		(iii)		Any five (×1) from: A. Proton pump pumps protons {into intermembrane space/ across membrane/ from Y to Z} (1) B. To create {proton / H ⁺ /electrochemical} gradient (1) C. {ATP synth(et)ase/ stalked particle} allows {protons / H ⁺ s} {through membrane/ down gradient} (1) D. {convert/combine} ADP (+ P _i) to ATP (1) E. correct reference to {electron carriers/ electron transport chain} (1) F. Ref to: Proton pump and ATP synth(et)ase (1)	3	2		5										